R2.2 How fast? The rate of chemical change

Energy profiles and molecularity (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- (R2.2.7) Construct and interpret energy profiles from kinetic data.
- (R2.2.8) Interpret the terms unimolecular, bimolecular and termolecular.

Energy profiles from kinetic data

Imagine yourself trekking up and down a scenic hilly area. Connecting with nature in this way can be very rewarding. Along the way you will experience routes that may offer considerable resistance and will require more input from you. Other times you may literally be running downhill.

Figure 1 maps the peaks and valleys for the Inca Trail.

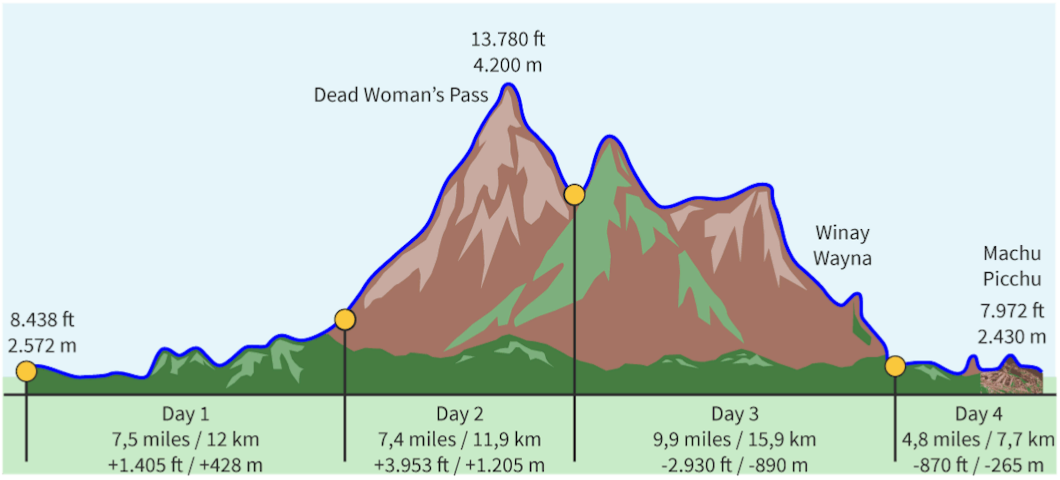

Figure 1. The Inca Trail map to Machu Picchu, in Peru, South America.

More information for figure 1

The image is an elevation profile of the Inca Trail to Machu Picchu. It shows a mountainous landscape divided into four segments, each representing a day of the hike.

- **Day 1**: Starts at 8,438 feet (2,572 meters) with a distance of 7.5 miles (12 km) and an elevation gain of 1,405 feet (428 meters).
- **Day 2**: The trail ascends sharply to Dead Woman's Pass, the highest point at 13,780 feet (4,200 meters). This segment covers 7.4 miles (11.9 km), gaining 3,953 feet (1,205 meters) in elevation.
- **Day 3**: The trail descends to Winay Wayna, covering 9.9 miles (15.9 km) and losing 2,930 feet (890 meters) in elevation.
- **Day 4**: Ends at Machu Picchu at 7,972 feet (2,430 meters), with a distance of 4.8 miles (7.7 km) and an elevation loss of 870 feet (265 meters).

Key locations such as Dead Woman's Pass and Winay Wayna are marked along the top of the profile. The blue line traces the trail over the mountains depicted in brown and green hues.

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Similarly, the path that reactions take can be represented to show the different steps and their related energy in an energy profile diagram. In section R2.2.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/reaction-mechanisms-hl-id-45509/) you learned how transition states and intermediates could be represented on energy profile diagrams. Energy profile diagrams can be used to show multi step reactions and can give useful information on the mechanism of the reaction. A multi-step reaction profile will be composed of a whole 'mountain range' with a series of dips and peaks, just like our Inca trail above. Take for example this reaction:

$$A_2 + 2BC \rightarrow B_2 + 2AC$$

The proposed mechanism for this reaction is:

Step 1:

$$A_2 + BC \rightarrow AB + AC$$

Step 2:

$$AB + BC \rightarrow B_2 + AC$$

Overall:

$$A_2 + 2BC \rightarrow B_2 + 2AC$$

This reaction can be represented on an energy profile diagram as shown in **Figure 2**.

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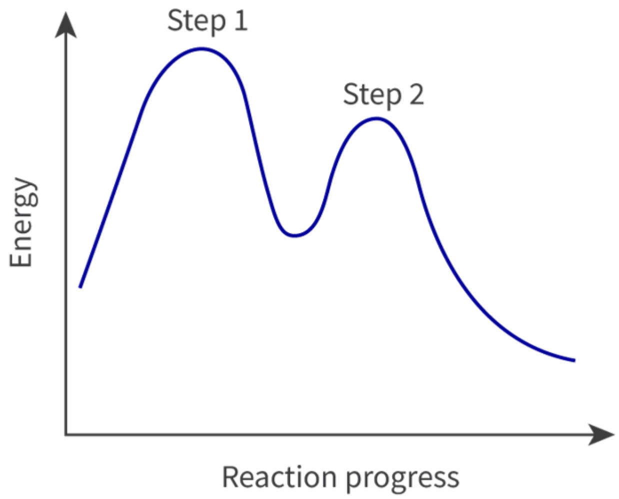

Figure 2. An energy profile for a multi-step reaction.

More information for figure 2

The image is a graph depicting an energy profile for a multi-step reaction. The X-axis is labeled 'Reaction progress,' and the Y-axis is labeled 'Energy.' The graph features a blue line that forms two peaks. The first peak is higher and labeled 'Step 1,' indicating an initial increase in energy. Following Step 1, the energy decreases before rising again to a second, smaller peak labeled 'Step 2.' After Step 2, the energy decreases continuously, suggesting the progression of the reaction. This graph provides insight into which step may be considered the rate-determining step based on the height of the peaks.

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Just by looking at the energy profile diagram in **Figure 2** can you tell which step is the rate determining step?

What information can be deduced from the energy profile diagram?

- We can see that it is a two-step reaction because there are two ‘peaks’ representing the transition states for each elementary step and one ‘dip’ representing the intermediate.
- The first transition state is at a higher energy level than the second, so the activation energy for the first step is higher than that for the second step. This means the first step is the slow step, or rate-determining step.
- The enthalpy change of the reaction is the distance between the reactants and products and can be marked on the diagram as shown in **Figure 3**.

The activation energies for the first step and second steps are E_{a1} and E_{a2} respectively. ΔH represents the enthalpy change of the reaction.

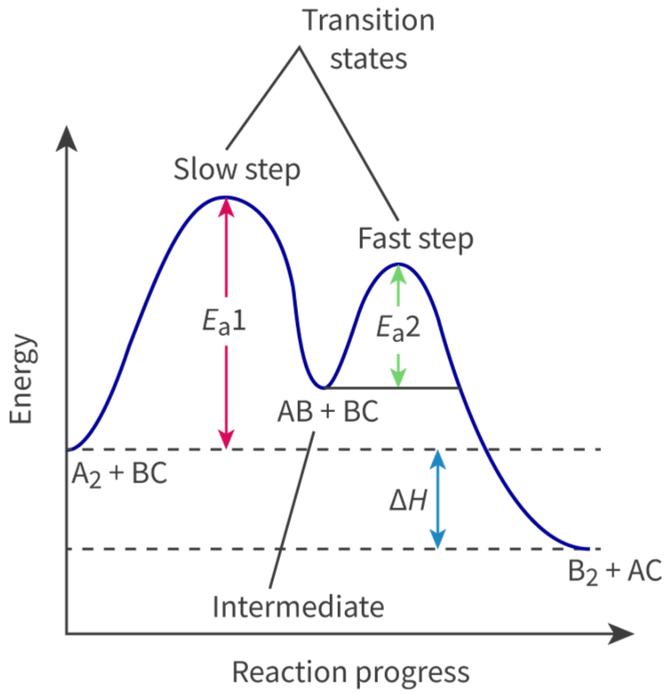

Figure 3. An energy profile diagram for a multi-step reaction showing the activation energies, E_{a1} and E_{a2} and the enthalpy change (ΔH) of the reaction.

More information for figure 3

The image displays an energy profile diagram for a multi-step chemical reaction. The X-axis represents 'Reaction progress' and spans from 'A2 + BC' to 'B2 + AC'. The Y-axis represents 'Energy'. Two peaks are evident, indicating transition states labeled as 'Slow step' and 'Fast step'. The first peak, representing the slow step, has a higher activation energy labeled 'Ea1'. The second, smaller peak represents the fast step with activation energy 'Ea2'. The intermediate stage is depicted between these peaks. A downward arrow labeled 'ΔH' indicates the enthalpy change from the top of the fast step to the product level. Key labels include 'AB + BC' within the intermediate area. The position of each transition state is marked with corresponding arrows indicating the direction of energy changes needed to proceed through each step.

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The overall activation energy of a multi-step reaction

It is very important to understand that a multi-step reaction may have different activation energies at each step, however the minimum energy required for the reaction to proceed must be equal to the energy needed to take the reactants to the transition state with the highest energy. **Figure 4** illustrates the activation energy for a multi-step reaction.

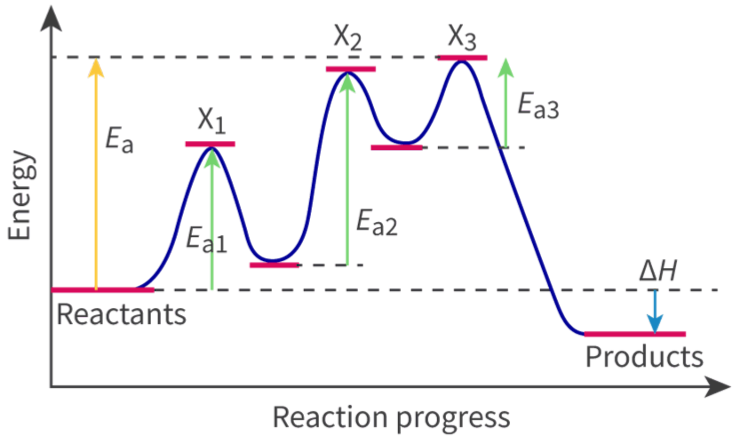

Figure 4. An energy profile diagram showing the activation energy of a multi-step reaction.

More information for figure 4

The image is a diagram illustrating the energy profile of a multi-step chemical reaction. The X-axis represents the reaction progress, ranging from Reactants on the left to Products on the right. The Y-axis represents the energy, with labels indicating different energy levels throughout the reaction.

There are three distinct energy peaks corresponding to three transition states labeled X1, X2, and X3. Each peak has an associated activation energy, labeled Ea1, Ea2, and Ea3, respectively. The overall activation energy, Ea, is represented by a yellow arrow pointing from the base energy level of the reactants to the top of the highest peak, X3. This step is identified as having the highest energy and is the rate-determining step.

The diagram shows energy valleys between each peak, representing intermediate steps in the reaction. The change in enthalpy, ΔH, is depicted by a blue arrow pointing downward from the reactants to the products.

The different components are interconnected by a blue curve outlining the energy path of the reaction, illustrating the energy changes at each step.

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The overall activation energy E_a is represented by the yellow arrow going from the reactants to the top of the third step which has the transition state at the highest energy X_3. The rate determining step is the second step with the highest activation energy E_{a2}.

You can practise labelling energy profile diagrams by putting the following labels into their correct place by using **Interactive 1**.

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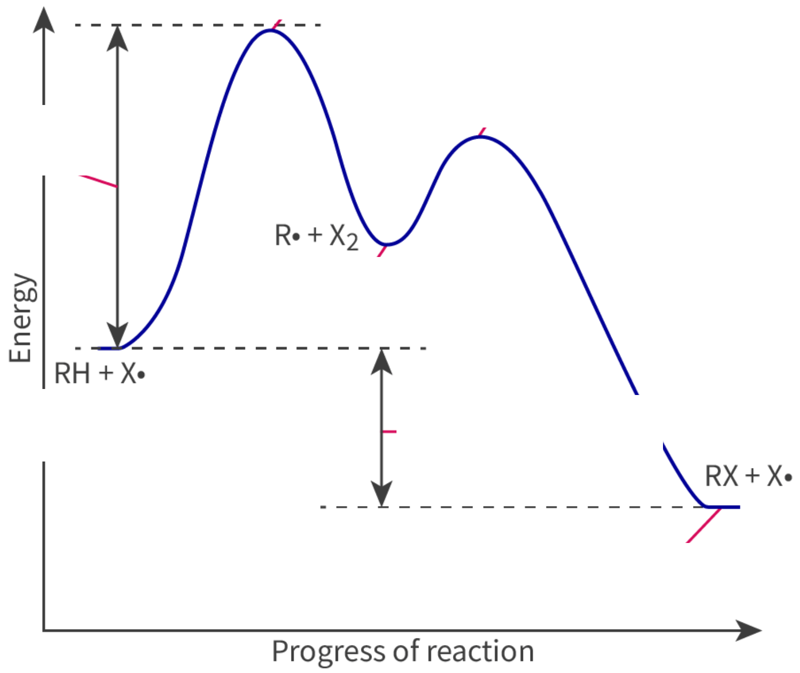

Fast step | Slow step | Enthalpy change

Starting materials | Intermediates | Activation energy | Products

Check

Interactive 1. Energy profile diagrams.

More information for interactive 1

An interactive energy profile diagram plots energy versus the progress of a reaction. A curve starts with the label $RH + X\cdot$, increases to form a large first peak, dips into a valley labelled $R\cdot + X_2$, increases again to form a small second peak, and finally descends to end at the label $RX + X\cdot$.
Users need to drag and drop the correct terminologies into the corresponding missing drop zones, which include:

$RH + X\cdot$
First peak
$R\cdot + X_2$
Second peak
$RX + X\cdot$

Distance between $RH + X\cdot$ and the first peak
Distance between $RH + X\cdot$ and $RX + X\cdot$
The drag-and-drop options provided are:
Fast step
Slow step
Enthalpy change
Starting materials
Intermediates
Activation energy
Products
This interactivity helps users to interpret an energy profile diagram for a multi-step chemical reaction and distinguish between fast and slow steps based on activation energy peaks.

Read below for the solution:

$RH + X\cdot$ - Starting materials
First peak - Slow step
$R\cdot + X_2$ - Intermediates
Second peak - Fast step
$RX + X\cdot$ - Products
Distance between $RH + X\cdot$ and the first peak - Activation energy
Distance between $RH + X\cdot$ and $RX + X\cdot$ - Enthalpy change

Worked example 1

Sketch the energy profile diagrams for the following reaction mechanisms. Label any intermediates on the energy profile diagram,

1. **Step 1, Fast:**
   A + B →AB
   Step 2, Slow:
   AB + A → A_2B
   Given that this reaction is endothermic.
2. **Step 1, Slow:**
   A + A→A_2
   Step 2, Fast:
   A_2 + B → A_2B
   Given that this reaction is exothermic.

1.

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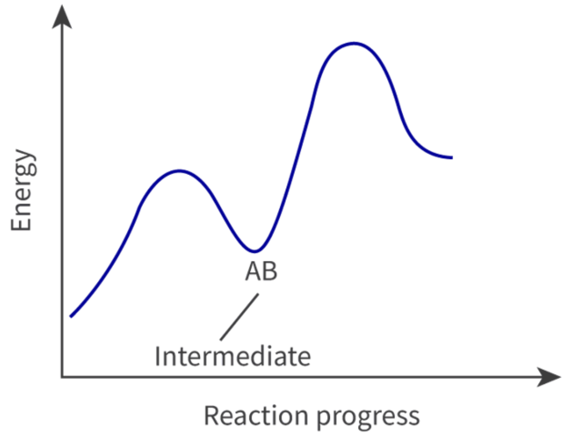

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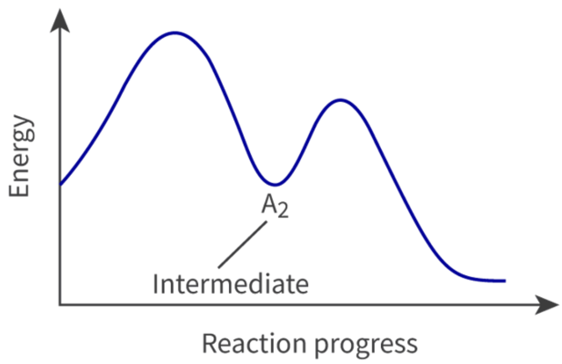


Molecularity of a reaction

For any elementary step in a reaction mechanism it is very important to know the number of reactant species involved. The term molecularity refers to the number of species reacting in an elementary step.

An elementary step where there is only one reactant is referred to as a unimolecular step or is said to have a molecularity of one.

Bimolecular refers to an elementary step with two reactant particles and the elementary step is said to have a molecularity of two.

A termolecular step, having a molecularity of three, is very rare. Why do you think this is so? It is because the probability of three reactant particles colliding with sufficient energy and in the correct orientation is very low.

Molecularity is always a whole number and is never greater than three.

For example, the following reaction is unimolecular, as there is only one reactant molecule (as seen directly from the stoichiometry of the reaction):

$$NH_4NO_2 \rightarrow N_2 + 2H_2O$$

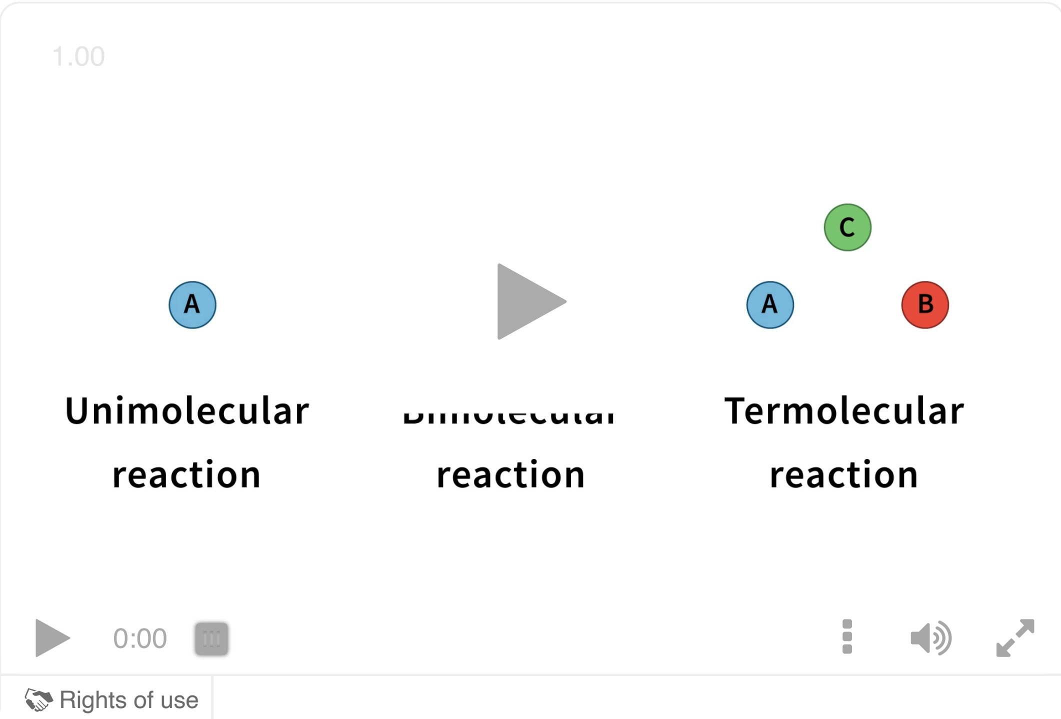
in contrast, this reaction:

$$2HI \rightarrow H_2 + I_2$$

is a bimolecular reaction as two HI molecules are colliding with the proper orientation and sufficient energy.

The following example is that of a termolecular reaction:

$$2NO + O_2 \rightarrow 2NO_2$$



Interactive 2. Molecularity.

More information for interactive 2

A video depicts three different reactions labelled unimolecular, bimolecular, and termolecular, demonstrating how products are formed in each case.
In the unimolecular reaction, a single reactant labelled A is converted into a product.
In the bimolecular reaction, two reactants, labelled A and B, react together to form a product.
In the termolecular reaction, three reactants, labelled A, B, and C, react together to form a product.

This video provides a comparison of unimolecular, bimolecular, and termolecular reactions, highlighting their differences in reactant participation and product formation.

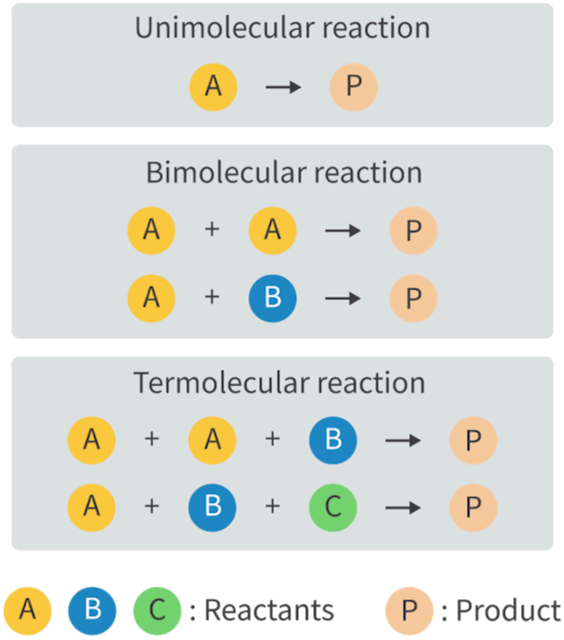


Figure 5. Molecularity in reactions.

More information for figure 5

The diagram illustrates different types of chemical reactions: unimolecular, bimolecular, and termolecular.

1. **Unimolecular Reaction:** This section shows a single reactant 'A' transforming into a product 'P'.
2. **Bimolecular Reaction:** This section contains two subsections:
3. The first shows two 'A' reactants combining to form a product 'P'.
4. The second shows one 'A' and one 'B' reactant combining to form a product 'P'.
5. **Termolecular Reaction:** This part includes two subsections:
6. The first shows two 'A' reactants and one 'B' reactant combining to form a product 'P'.
7. The second shows one 'A', one 'B', and one 'C' reactant combining to form a product 'P'.

Below the reactions, a legend explains the color-coded symbols: 'A', 'B', and 'C' are reactants, while 'P' is the resulting product. The diagram uses arrows to indicate the direction of each reaction.

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Worked example 2

One of the proposed mechanisms for the thermal decomposition of nitryl chloride is:

Step 1 (slow step):

$$NO_2Cl \rightarrow NO_2 + Cl$$

Step 2 (fast step):

$$NO_2Cl + Cl \rightarrow NO_2 + Cl_2$$

Overall:

$$2NO_2Cl \rightarrow 2NO_2 + Cl_2$$

Identify the molecularity of the elementary steps in this mechanism.

Step 1 is unimolecular

Step 2 is bimolecular

Activity

- **IB learner profile attribute:**
  - Knowledgeable
  - Inquirer
  - Thinker
  - Reflective
- **Approaches to learning:**
  - Thinking skills — Providing a reasoned argument to support a conclusion, Applying key ideas and facts in new contexts
  - Communication skills — Using terminologies, symbols and communication conventions consistently and correctly
  - Social skills — Working collaboratively to achieve a common goal
- **Time required to complete activity:** 15—20 minutes
- **Activity type:** Group activity

Work in groups of three to make an energy profile diagram for the following two-step exothermic reaction:

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Step 1:

$$A + B \rightarrow AB$$

Step 2:

$$AB + A \rightarrow A_2B$$

Each person in the group will make a separate energy profile diagram. One showing the first step as the slow step and one where the second step is the slow step.

1. Compare the profile diagrams in each group and deduce the overall equation.
2. Identify the molecularity of each elementary step.